👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 1 Measuring Tools (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
His readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as his worksheets for that lesson
- Quick check: score, then scale to his test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Micrometer Handout**: The handout is a scanned image with no text; our notes teach the same skill.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

📏 Measuring Tools: Guided Notes

Small Engines Overhaul · Unit 1 · Level 3 (closer to the class work) · goes with the Measurement Lab, Micrometer Handout and cylinder bore video

Name: _____

Use the class slides and your reading. Key words:

precision *dial caliper* *micrometer* *thimble* *sleeve* *feeler gauge* *specification* *diameter* *cylinder bore* *taper*

A. Tools

B. Reading

2. **Add the readings**

> Caliper: bar 0.400 + dial 0.032 = 0.432
> Micrometer: sleeve 0.750 + thimble 0.011 = 0.761

a) Caliper: bar 0.600 + dial 0.018 _____

b) Micrometer: sleeve 0.875 + thimble 0.004 _____

c) Caliper: bar 1.200 + dial 0.050 _____

C. Specs

3. A journal spec is 0.873-0.875. You measure 0.869. What do you conclude?

_____

_____

4. The bore is measured at the top, middle and bottom to check for _____.

📏 Measuring Tools: Reading

Small Engines Overhaul · Unit 1 · Level 3 (closer to the class work) · goes with the Measurement Lab, Micrometer Handout and cylinder bore video

Name: _____

Precision Measuring in the Engine Shop

Engine parts are made to very exact sizes. Mechanics measure them to one thousandth of an inch, written 0.001. A sheet of paper is about three thousandths thick, so these tools measure very small differences.

A dial caliper has jaws that slide along a bar. The outside jaws measure the outside of a part, the inside jaws measure a hole, and the depth rod measures how deep a hole is. To read it, read the inches and tenths on the bar, then add the thousandths on the dial.

A micrometer is even more exact. You turn the thimble until the spindle touches the part, using the ratchet so you do not squeeze too hard. Read the sleeve (each line is 0.025) and add the thimble (each line is 0.001).

A feeler gauge is a set of thin blades with known thicknesses. It measures small gaps like spark plug gap, ring end gap and valve clearance. The repair manual gives the specification for every part. If your measurement is outside the spec, the part is worn and may need to be replaced.

Check your reading

1. What is a thousandth of an inch, written as a decimal?

2. How do you read a dial caliper?

3. What does it mean if a part measures outside the spec?

📏 Measurement Stations: Lab Sheet

Small Engines Overhaul · Unit 1 · Level 3 · goes with Measurement Lab Sheet

Name: _____

Question: Are the station parts within the manufacturer's specification?

Materials

dial caliper, micrometer, feeler gauges, ruler, station parts, spec sheet

⚠️ Safety

- Safety glasses on
- Handle precision tools gently and return them to cases

Prediction

Which part do you predict will be most worn, and why?

_____

_____

Procedure

1. Clean each part.
2. Measure the rod diameter and length with the caliper.
3. Measure 3 crankshaft journals with the micrometer.
4. Find the feeler blade for the sample gap.
5. Record measured vs spec; decide in or out of spec.

Data

Part	Spec	Measured	In spec?
Rod diameter			
Magneto journal			
Crankpin			
PTO journal			

Analysis (Claim, Evidence, Reasoning)

1. Claim: is this crankshaft reusable? Evidence: your measurements vs spec. Reasoning: what wear does to the engine.

📏 Measuring Tools: Quick Check

Small Engines Overhaul · Unit 1 · Level 3 (closer to the class work) · goes with the Measurement Lab, Micrometer Handout and cylinder bore video

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Which tool measures a hole's depth?

A. Feeler gauge | B. Ruler only | C. Dial caliper

2. Each thimble line on a micrometer equals…

A. 0.001 inch | B. 0.025 inch | C. 0.1 inch

3. A feeler gauge is used for…

A. Weighing parts | B. Measuring fuel | C. Gaps like spark plug gap

4. Caliper bar 0.300 + dial 0.047 =

A. 0.347 | B. 0.3047 | C. 3.47

5. Why do mechanics compare measurements to the spec?

6. Describe how to measure a cylinder bore.

📏 Measuring Tools: Guided Notes · ANSWER KEY

Small Engines Overhaul · Unit 1 · Level 3 (closer to the class work) · goes with the Measurement Lab, Micrometer Handout and cylinder bore video

Name: _____

Use the class slides and your reading. Key words:

precision | *dial caliper* | *micrometer* | *thimble* | *sleeve* | *feeler gauge* | *specification* | *diameter* | *cylinder bore* | *taper*

A. Tools

B. Reading

2. **Add the readings**

> Caliper: bar 0.400 + dial 0.032 = 0.432
> Micrometer: sleeve 0.750 + thimble 0.011 = 0.761

a) Caliper: bar 0.600 + dial 0.018 **0.618**

b) Micrometer: sleeve 0.875 + thimble 0.004 **0.879**

c) Caliper: bar 1.200 + dial 0.050 **1.250**

C. Specs

3. A journal spec is 0.873-0.875. You measure 0.869. What do you conclude?
Below spec, so it is worn; replace or check with the teacher.

4. The bore is measured at the top, middle and bottom to check for **taper** .

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

📏 Measurement Stations: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 1 · Level 3 · goes with Measurement Lab Sheet

Name: _____

Question: Are the station parts within the manufacturer's specification?

Materials

dial caliper, micrometer, feeler gauges, ruler, station parts, spec sheet

⚠️ Safety

- Safety glasses on
- Handle precision tools gently and return them to cases

Prediction

Which part do you predict will be most worn, and why?
Answers vary; crankpin journals often wear most.

Procedure

1. Clean each part.
2. Measure the rod diameter and length with the caliper.
3. Measure 3 crankshaft journals with the micrometer.
4. Find the feeler blade for the sample gap.
5. Record measured vs spec; decide in or out of spec.

Data

Part	Spec	Measured	In spec?
Rod diameter	**0.500**	**0.499**	**yes**
Magneto journal	**0.873**	**0.872**	**yes**
Crankpin	**0.873**	**0.868**	**no (worn)**
PTO journal	**1.001**	**1.000**	**yes**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: is this crankshaft reusable? Evidence: your measurements vs spec. Reasoning: what wear does to the engine.
Example: crankpin is 0.005 under spec, so it is worn; replace it or it will knock and lose oil pressure.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

📏 Measuring Tools: Quick Check · ANSWER KEY

Small Engines Overhaul · Unit 1 · Level 3 (closer to the class work) · goes with the Measurement Lab, Micrometer Handout and cylinder bore video

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Which tool measures a hole's depth?

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2. Each thimble line on a micrometer equals…

A. 0.001 inch | B. 0.025 inch | C. 0.1 inch

3. A feeler gauge is used for…

A. Weighing parts | B. Measuring fuel | **C. Gaps like spark plug gap**

4. Caliper bar 0.300 + dial 0.047 =

A. 0.347 | B. 0.3047 | C. 3.47

5. Why do mechanics compare measurements to the spec?

To find worn parts that must be replaced so the engine runs right.

6. Describe how to measure a cylinder bore.

Clean it; measure at top, middle and bottom in two directions; compare to spec for wear, taper, out-of-round.

🔑 *Written by our team from his unit calendar, worksheets and review games, please check.*